EXP 2

Editor

ADCexp2.m

/MATLAB Drive/ADCexp2.m

```
1   clc;
2   clear;
3   close all;
4
5   Am = 1;
6   fm = 500;
7   Ac = 5;
8   fc = 5000;
9
10  fs = 20*fc;
11  t = 0:1/fs:4/fm;
12
13  % Message signal
14  m = Am*sin(2*pi*fm*t);
15
16  % Carrier signal
17  c = Ac*sin(2*pi*fc*t);
18
19  % Modulation index
20  mu = Am/Ac;
21
22  % AM signal
23  am = Ac*(1 + mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
24
```

```
% Demodulation using Hilbert transform
dem = abs(hilbert(am));
dem = dem - mean(dem);

% FFT
N = length(am);
f = linspace(-fs/2, fs/2, N);
AM_FFT = abs(fftshift(fft(am)))/N;

% Sideband frequencies
USB = fc + fm;
LSB = fc - fm;

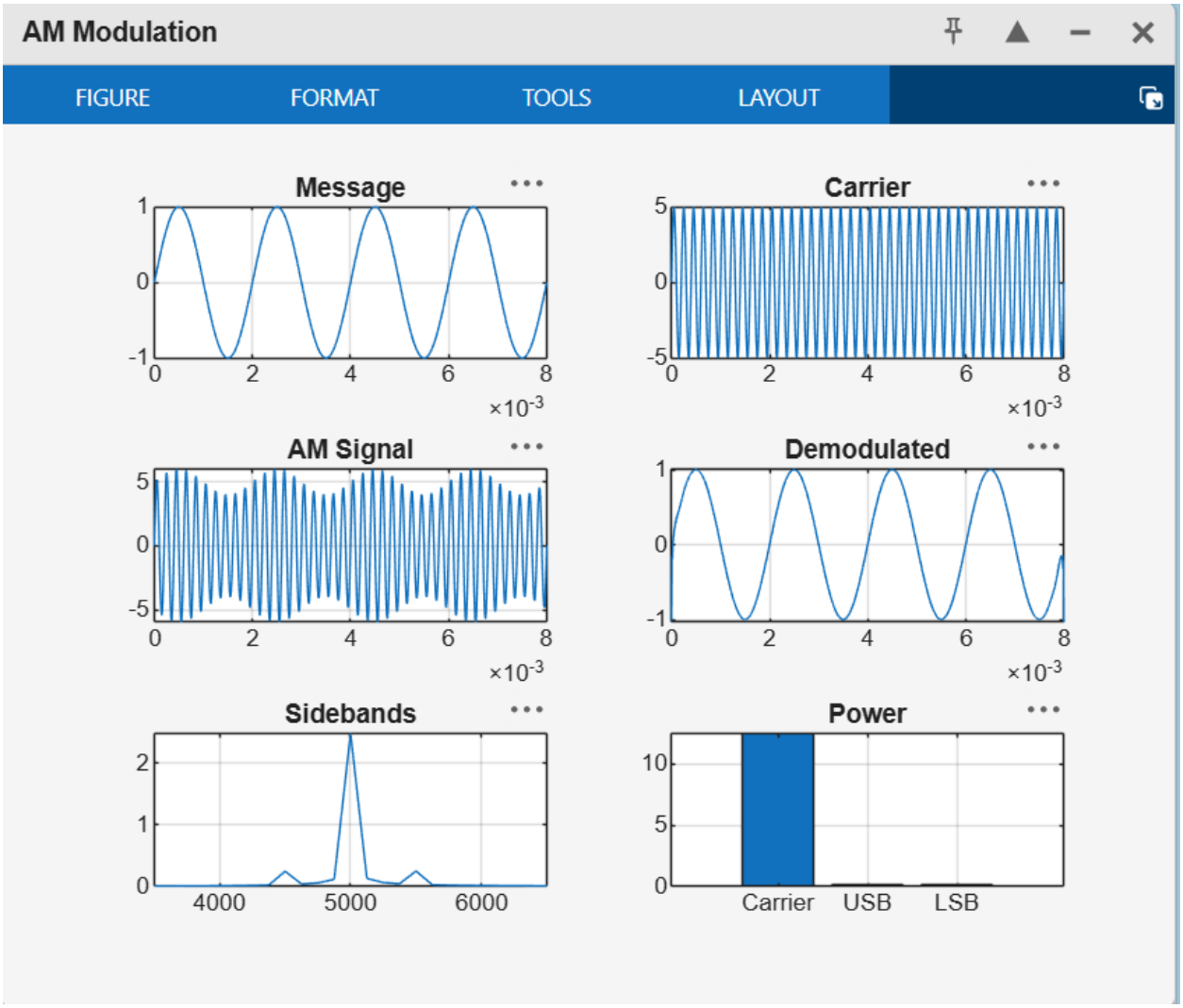
% Power calculations
Pc = Ac^2/2;
Psb = (mu^2 * Pc)/4;

% Plotting
figure('Name','AM Modulation','NumberTitle','off');

subplot(3,2,1);
plot(t,m);
title('Message');
grid on;
```

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```
50      subplot(3,2,2);
51      plot(t,c);
52      title('Carrier');
53      grid on;
54
55      subplot(3,2,3);
56      plot(t,am);
57      title('AM Signal');
58      grid on;
59
60      subplot(3,2,4);
61      plot(t,dem);
62      title('Demodulated');
63      grid on;
64
65      subplot(3,2,5);
66      plot(f,AM_FFT);
67      xlim([fc-3*fm fc+3*fm]);
68      title('Sidebands');
69      grid on;
70
71      subplot(3,2,6);
72      bar([Pc Psb Psb]);
73      set(gca,'XTickLabel',{'Carrier','USB','LSB'});
```


AM Modulation
FIGURE
FORMAT
TOOLS
LAYOUT
Message
1
0
-1
0
2
4
6
8
×10-3
Carrier
5
0
-5
0
2
4
6
8
×10-3
AM Signal
5
0
-5
0
2
4
6
8
×10-3
Demodulated
1
0
-1
0
2
4
6
8
×10-3
Sidebands
2
1
0
4000
5000
6000
Power
10
5
0
Carrier
USB
LSB